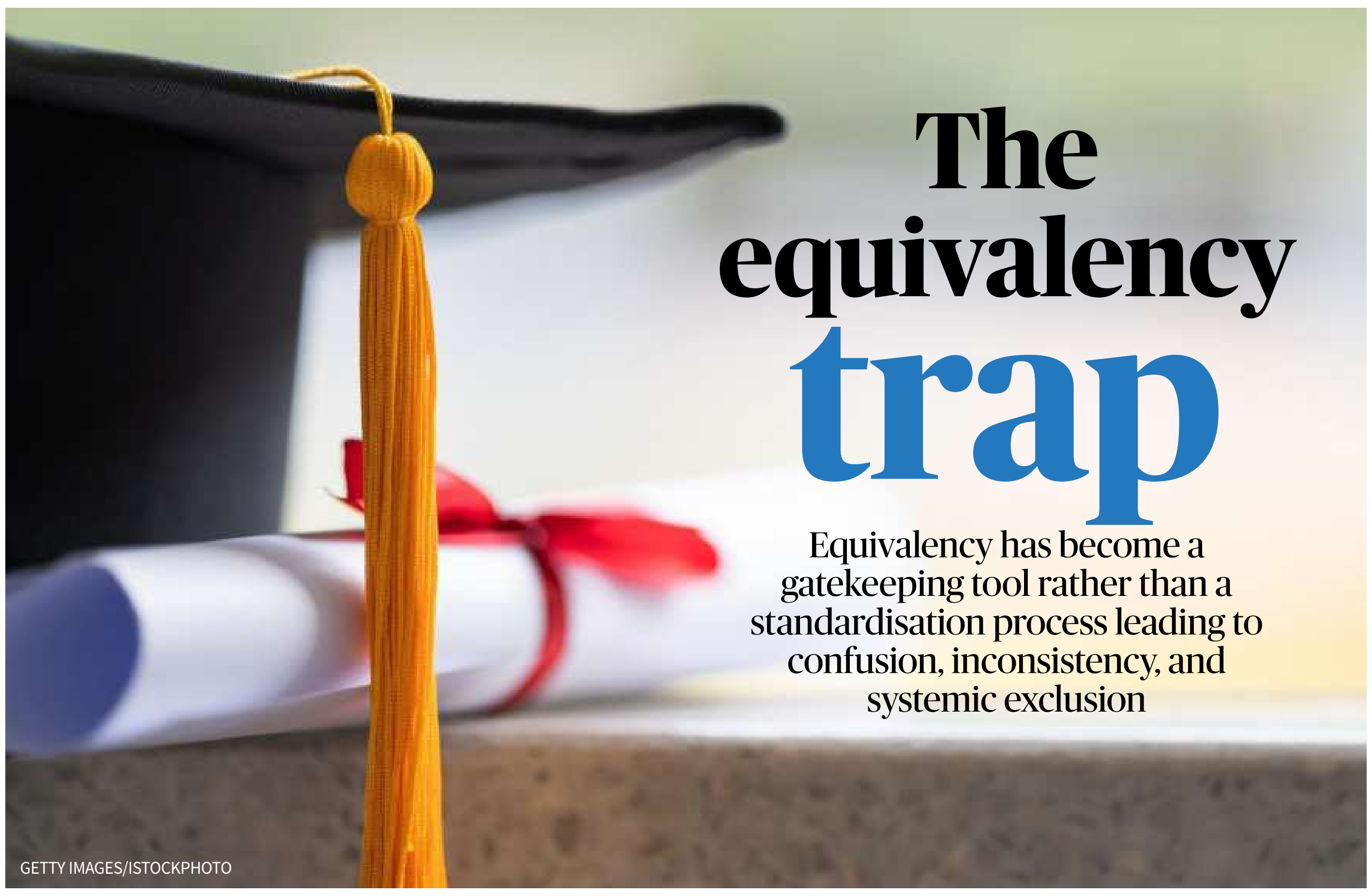

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The equivalency trap

Equivalency has become a gatekeeping tool rather than a standardisation process leading to confusion, inconsistency, and systemic exclusion

Sanil Raj Johnson

As higher education in India becomes more flexible and multidisciplinary, the continued requirement of the Equivalency Certificate – that students need to submit when applying at different universities – remains a barrier to student mobility. What initially appears to be a simple academic verification has, over time, transformed into a complex bureaucratic process.

At its core, the system is meant to ensure that a student's prior degree meets the receiving university's academic standards. However, without a national policy and a clear mechanism, universities have started to see equivalency as a gatekeeping tool rather than a standardisation process. This has led to confusion, inconsistency, and, more worryingly, systemic exclusion.

Long process

To begin with, securing equivalency requires students to gather an exhaustive set of documents: detailed syllabi for every semester, course scheme and regulations, the programme structure and duration, mode of examination and even evidence such as the institution's NAAC grade and NIRF ranking. The university's Registrar must authenticate these, a process that can take weeks. After these documents are submitted, their application moves through different layers of scrutiny, typically beginning with the relevant Board of Studies and ending with the Academic Council.

This long process does not guarantee approval. Many universities insist on an unwritten rule that the syllabus submitted must match at least 60% of their own. This has no basis in the University Grants Commission's regulations, yet is widely and rigidly enforced. It is also deeply flawed because no two universities design their programmes with uniformity in mind. Autonomous colleges, state universities, and institutions across different States develop their syllabi independently based on regional needs, faculty expertise, and institutional priorities. Expecting a significant percentage of similarity between such diverse curricular frameworks is unrealistic.

Double and triple majors

The situation becomes even more complicated when students graduate from institutions offering double-major or triple-major degrees. These programmes have expanded steadily in India over the last decade, long before the introduction of the Four-Year Undergraduate Programme. Their purpose is to broaden learning and enable students to pursue postgraduate studies in any one of their major subjects. But when students apply for higher studies elsewhere, they are forced to prove equivalency in one of the subjects they studied, even when the postgraduate programme does not require subject-specific eligibility. A student with a triple-major degree in Economics, Political Science and Journalism may be denied admission to a Master of Social Work programme simply because none of the subjects matches the 60% similarity criterion, even though MSW programmes typically accept graduates from any discipline.

A recent case illustrates the system's irrationality. A student who completed a triple-major degree in Psychology, Zoology, and Botany from a university in Karnataka sought admission to an M.A. English programme at a university in Kerala. According to the regulations, a graduate with the required marks and performance in the entrance examination was eligible, regardless of the subjects studied at the undergraduate level. The student met these requirements and qualified through the entrance test.

However, because he had studied three majors, the institute asked him to secure equivalency in at least one. After reviewing his syllabi, he found that Botany had the closest overlap and applied for equivalency in that subject. Unfortunately, the Board decided that what he had studied in Karnataka did not align sufficiently with what was taught in Kerala and rejected the request. As a result, he was denied admission, despite Botany having no relevance to the course. Thus, equivalency became a procedural hurdle that blocked a student who had already demonstrated his merit.

Need for one policy

Cases such as these are not anomalies. With more than four crore students enrolled in India's higher education institutions and increasing numbers choosing to pursue degrees outside their home States, the demand for equivalency certificates is rising sharply.

India needs a transparent national policy on equivalency to support student mobility and prevent universities from enforcing arbitrary criteria. This must abandon outdated subject-wise equivalency requirements for postgraduate programmes that already accept graduates from any discipline. It should also recognise multidisciplinary degrees without forcing students to approach unrelated Boards of Studies and shift the focus from syllabus matching to outcome-based measures that reflect modern academic standards.

The country is at a pivotal moment in its educational reform. While the National Education Policy promotes flexibility, interdisciplinarity, and student-centred learning, the current equivalency system remains one of the most regressive elements of university administration.

Views are personal.

The writer is a Professor of English and Dean of Student Affairs, Sahrdaya College of Advanced Studies (Autonomous), Thrissur, Kerala. Email: saniljohn03@gmail.com

SCHOLARSHIPS

IIT-Jodhpur Internship Programme
Offered by IIT-Jodhpur.
Eligibility: Those who have completed fifth semester of B.Tech./B.S. or first semester of M.Sc. with minimum CGPA of 7 or above.
Rewards: ₹6,000 monthly for two months
Application: Online
Deadline: February 28
www.b4s.in/edge/IJRI1

University of Glasgow India Excellence Award
Offered by the University of Glasgow, the U.K..
Eligibility: Indian passport holders who want to pursue a full-time postgraduate taught Master's in any discipline in September and demonstrate academic excellence equivalent to UK First Class Honours, and meet financial requirements .
Rewards: £12,500 tuition fee waiver.
Application: Online
Deadline: March 1
www.b4s.in/edge/UGIA1

UK International Undergraduate Excellence Scholarships
Offered by the University of Nottingham, the U.K.
Eligibility: International, fee-paying students who have AAA grades or its international equivalent and hold an offer to begin a full-time UG degree.
Rewards: Up to £2,000
Application: Online
Deadline: March 11
www.b4s.in/edge/NIUE1

Courtesy: buddy4study.com

Feeling stuck?

Uncertain about your career options? Low on self-confidence? This column may help

OFF THE EDGE
Nandini Raman

I am a B.Tech (Electronics and Instrumentation) graduate and a Red Hat Certified Engineer working as a Linux Technical Support Engineer. Rotating three shifts has made the role exhausting and unfulfilling. What career direction can I consider? Jayaram

Dear Jayaram,

To move into roles that offer better work-life balance and significantly higher pay, you may need to certify and upgrade your current skillset, which four to six months of focused learning should do.

When applying, look for "Product-based companies" or "Internal IT" rather than "Managed Service Providers" (MSPs) because they require shifts to support global clients.

In contrast, product companies (such as a FinTech startup or a SaaS company) have their engineering teams on a fixed daytime schedule, with only occasional "on-call" rotations.

Based on your RHCE background, consider roles like a DevOps Engineer. You will need to learn Docker and Kubernetes (containerisation) and Jenkins or GitLab CI (CI/CD pipelines), and fill the skill gap quickly.

Alternatively, consider a Cloud Infrastructure Engineer role. You will need to get a Solutions Architect certification in AWS or Azure. Learn Terraform for Infrastructure as Code (IaC). A Site Reliability Engineer (SRE) uses software engineering to solve operational problems, which adds a stronger coding layer to operations. Move beyond Bash and become proficient in Python or Go.

I have done B.Des. Footwear Designing. I am not getting a job or able to get into a PG course. I am not good at exams. How can I make a fresh start? Hiba

Dear Hiba,

The design industry is one of the few fields where your portfolio matters more than your mark sheet. Pivot to high-demand segments such as sneaker customisation. Customise sneakers by painting, embroidery, and deconstruction and create dedicated social media pages to showcase them. Your product and industrial design skills are transferable.

Your skills in material science, ergonomics, and 3D modelling (CAD) can get you a "Junior Product Designer" role in consumer goods, luggage, or wearable tech companies.

Build a portfolio that clearly demonstrates your work. Employers in design only care about what you can make. Create a few high-quality projects, exhibit your rough sketches, your material mood boards, and your technical 'tech packs' to show that you understand the process.

Network on LinkedIn. Find "Design Leads" at companies such as Bata, Liberty, and Metro, or at startups like Neeman's, and request to showcase your portfolio.

If you want to study more, look out for skill-based diplomas (usually three to six months) that offer direct placement help. Check out the UI/UX Design and Visual Merchandising and Retail space. Also, see if you want to create a micro entrepreneurial brand of your own. Start a "Direct-to-Consumer" (D2C) brand on a small scale with your choice of footwear. Find local artisans to work to your design and ensure your quality checks are perfect.

I graduated in 2023 (Civil Engineering). I have been preparing for the UPSC CSE for the last couple of years. Since there is a three-year gap, how can I get a job in a civil engineering company if I fail to clear the UPSC on my fourth attempt? Sandeep

Dear Sandeep,

You will need to quickly update your skills and your CV to become job-ready. Work on key software skills gaps that ensure specialised roles, such as a certification in Master tools STAAD.Pro, ETABS, and advanced Excel for Design/Structural Roles, MS Project or Primavera P6 for Planning roles, and learn Autodesk Revit.

Companies now mandate Building Information Modelling (BIM) for large infrastructure projects. Look for training institutes that bridge the gap between college and site reality and offer short-term courses.

In the interviews, do not be apologetic about the gap in your CV and focus on showcasing the realignment of getting certified in BIM and AutoCAD. Apply to smaller design Firms that will look at whether you can deliver working drawings in AutoCAD immediately.

Companies in solar energy, EV charging station construction, warehouse development, third-party inspection (TPI) agencies hire civil engineers for quality audits.

I am doing my PG in Food Science and Nutrition (Community Science). What are my career options in India and abroad? Will a PhD enhance my prospects? Veda

Dear Veda,

In India, you can explore options such as a food analyst with FSSAI and other government bodies. Across the FMCG Industry, you can consider roles such as product development or quality assurance.

Another option is a public health nutritionist for community-level interventions across public health organisations and NGOs.

You can also explore clinical research across hospitals that hire PG graduates to assist in Nutraceutical R&D.

International options include organisations such as the WHO or FAO, and global food tech companies that have global R&D hubs focussing on clean-label and sustainable food.

High-demand markets include the U.S., Canada, and West Asia, which are seeing a strong demand for regulatory affairs and therapeutic nutritionists.

A Ph.D. is not necessary for a job in a factory or a clinic, but is transformative for your long-term goals and career path, especially if you decide to get into academia or teaching.

Disclaimer: This column is merely a guiding voice and provides advice and suggestions on education and careers.

The writer is a practising counsellor and a trainer. Send your questions to eduplus.thehindu@gmail.com with the subject line Off the Edge.

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Beyond bricks and mortar

The future of Civil Engineering will be shaped by smart technologies, sustainable building practices, and digital transformation

Rafat Siddique

Out of 1.5 million Engineering graduates every year, around 300,000 are from Civil Engineering. Currently, they are not just builders of passive structures but have become the vital problem-solvers of unavoidable urban disasters. From responding to floods caused by climate change and the functionality of smart transportation networks to ensuring safe drinking water, the profession is in a dynamic stage of change. This new paradigm demands Civil Engineering education to become more interdisciplinary, industry-focused, and technologically integrated.

Civil Engineering is the application of physical and scientific principles to perform the design, development, and maintenance of built and naturally built environments and is a foundational field that includes all critical infrastructure, including structural assessment (buildings, bridges), transportation planning, geo-technical studies, and environmental sustainable systems (water management, energy, pollution). The future of civil engineering requires graduates to combine core domains with data science, policy, and finance.

The sector is expected to grow at a CAGR of 7.10% from 2025 to 2033, driven by infrastructure development and technological advances. With this growth comes a need to significantly rethink the foundational technical skills. It requires graduates to learn technologically advanced tools such as Building Information Modelling (BIM), AutoCAD Civil 3D, or Revit that build proficiency. In addition, technologies such as AI are also entering the field. AI-generated predictive analytics are essential to assess the condition of infrastructure. Data interpretation and machine learning are specialised skills that separate modern civil engineers from traditional ones.

Integrated knowledge

Modern projects are complex, requiring civil engineers to integrate knowledge across various disciplines. Civil Engineering education has, therefore, become interdisciplinary, leading to the development of more innovative, sustainable, and efficient solutions. For example, the construction industry produces 20-30% of all the air pollution that we generate. Civil engineers must be aware of new materials such as ultra-high-performance concrete, self-healing materials, and recycled materials to address the environmental impact of construction.

Additionally, through interdisciplinary disciplines such as urban planning, data analytics, and even psychology, civil engineers can develop comprehensive and effective solutions for more complex projects. Psychology allows civil engineers to use rigorous scientific techniques to understand human capabilities and limitations, which is important for mitigating failures (e.g., improper use of a structure, such as excessive occupancy) and injuries in work contexts.

As a result, industry-relevant internships and professional experiences are not optional, as they establish a critical connection between the classroom and workplace. Through hands-on experience, students engage, explore, and refine their theoretical understanding into productive knowledge and experience.

Soft skills

In addition to technical proficiency, the development of soft skills is vital for career advancement and successful project delivery. Research indicates that the two most popular skills are communication and collaboration to work with non-technical stakeholders such as clients, government regulators, and the public. Additionally, project management skills are necessary to coordinate complicated logistics, scheduling, and risk when working on major infrastructure projects. Finally, leadership and ethics are required to make informed decisions about public safety, policies, and environmental implications, affirming the societal responsibility of civil engineers.

Engineering is pivotal in India's evolution, from constructing large infrastructure in the post-Independence era to driving the digital economy today. However, the demands on engineers are changing more quickly than most institutions can adapt. The future of Civil Engineering will be shaped by smart technologies, sustainable building practices, and digital transformation to build climate-resilient infrastructure, smart cities and green transport systems. This means that institutions need to rethink how they teach and learn.

The writer is a Senior Professor of Civil Engineering at Thapar Institute of Engineering and Technology

Avik Chattopadhyay

Taught, not used

First sharpen students' cognitive skills and hone their creative quotient. AI can come later

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Could I have used AI to write what follows? Yes. Should I have used AI to write what follows? I'd better not. For if I had, then what you will be reading need not be what I really believe in. It might well be what anybody else could espouse, bereft of experience, emotion and enterprise. Most importantly, it would be the outcome of all that has been digitally tracked, traced, and captured till one minute before. Not of how the mind thinks right now, in a certain context and with a certain purpose.

AI is necessary for all that we currently do and wish to do in the decades ahead, till some other interface betters it. But it is an interface. One of the means. A facilitator. Not the end. It is like speech and writing. They are the tools we have used in our process of evolution, not the product.

Not yet time

So, should AI be used in Management education? No. Why not? Because it is not yet time for students of Management to obsess with artificial tools before they develop their own capabilities and skills.

But it is already a part of everyday life. So, should students not be prepared accordingly? Driving, too, is a part of everyday life, but do you allow anyone before a certain age to indulge in the same?

It is not so much to do with age, but with a level of internal development. Social scientists say that it all happens with age. So, my opinion is that AI should not be used in management education.

But used is different from being taught. So, should AI be taught as part of management education? For sure. The future manager or entrepreneur should be aware of all tools that can improve the process and productivity. Just like we have sex education in school and workshops on traffic rules.

Management education is also going through soul-searching now. The corporate ecosystem is questioning its current form, content, and purpose. Today, it is largely classroom-based, structured and focused on creating managers rather than entrepreneurs. One does not learn intuition and individualism. Creativity is at best taught, not necessarily encouraged or practised. In this state of flux, incorporating yet another "structure" in the form of AI as a tool to be used for education will further complicate the soul-searching.

Teachers and students

Also, when we talk of education, it applies to teachers as much as students. I strongly believe that no teacher should use AI to craft syllabi, lessons, and classes. That defeats the entire purpose of demonstrating to the students the power of their own thinking and individualism. Again, I am not against teaching what is happening in the world of AI. Students need to know about Agentic, Ambient, and Neurosymbolic and the differences between them, but they should not be used at this stage.

Some may say I am being a Luddite and against technology and progress. They may argue that calculators were finally allowed in class. Agreed, and they should have been once they were accepted as a medium to allow better output, not as the output itself. It takes a generation or two for that realisation to happen, and we need to give it the required time.

Management education should not be about learning quantitative techniques, formulae, laws and sections in an act. That was yesterday. Tomorrow has to be more about cognitive fluency, the ability to plan, prepare, execute and solve issues or problems using a mix of the structured and the instinctive. That makes every outcome totally bespoke and individualistic. Recommending ready-made solutions, under the garb of "not reinventing the wheel" and "compressing gestation periods", has actually dumbed down Management education. The fact that we now cannot read long formats and prefer video over written content is an outcome of an artificial curriculum structure resting on classroom examinations, synthetic role plays, and an archaic evaluation system instead of one based on multiple internships, collaborative projects with the industry, and evaluation by impact.

Using AI in such a framework will make Management education obsolete. Let us focus on sharpening cognitive skills and honing the creative quotient. Artificial Intelligence can be handled later.

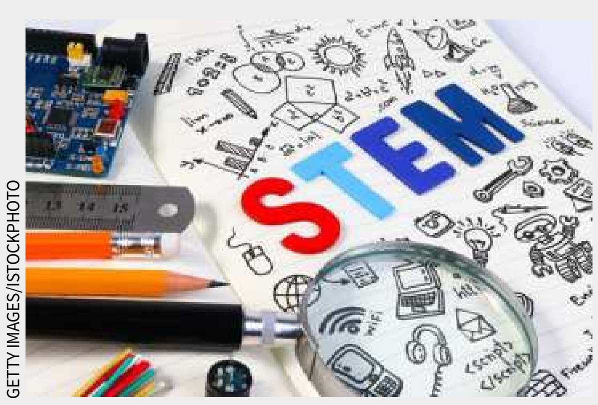
The writer is the chairperson of the XLRI Centre for Automobile Design and Management (XADM) and founder of Indian School for Design of Automobiles (INDEA) at XLRI-Delhi NCR.

Summer strategy

A look at leading STEM programmes in India and abroad during the long break

CAREER CUES

Richa Dwivedi Saklani



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As high school students begin to consider college applications, summer breaks are spent either exploring or building on an area of interest. Let's take a look at leading STEM programmes in India and abroad.

India

Ashoka Young Scholars Programme (YSP)

Ashoka University, Sonipat

A structured pre-college programme offering university-level exposure through interactive courses. Two tracks —Innovators on Science and Tech and Tech, Data, and Computer Science – encourage scientific thinking along with other multidisciplinary courses. Best suited for students seeking academic depth, discussion-based learning, and exposure to a top Indian university environment.

Dates: May 18 to 29 and June 1 to 12

Deadline: Early/scholarship consideration by March 8, final deadline in late April

Fees: ₹85,000 + GST (financial aid available)

YTS+

Plaksha University, Mohali

Earlier called Young Technology Scholars, YTS+ is a hands-on STEM programme focused on Technology, Engineering, AI, and Design Thinking. Students work in teams on real-world problems under faculty mentorship. Best suited for those interested in Engineering, Computer Science, and Applied Technology.

Dates: May 24 to June 7

Deadline: Round 1 is February 22, Round 2 is March 29

Fees: ₹105,000 + GST (scholarships available)

Symbiosis Summer School

Symbiosis International University, Pune

A two-week structured exploratory programme offering various academic disciplines, campus exposure, and certification. Meant for students interested in understanding their unique strengths through Symbiosis' Strong Interest Inventory and getting clarity and confidence in their academic choices.

Dates: May 18 to 30 and June 1 to 13

Deadline: Usually March

Fees: ₹45,000 (varies depending on the track)

Global programmes

Wharton Global Youth Programme

MTSI, the U.S.

This highly rigorous and selective course introduces students to the knowledge necessary to develop concepts and skills linking Management and Technology and is taught by professors from Wharton and Penn Engineering. This is best suited for students curious about techpreneurship.

Dates: July 5 to 25

Deadline: March 25

Fees: $9,000

Berkeley M.E.T. Innovation Academy

University of California-Berkeley, the U.S.

Berkeley METia is a hands-on summer programme that explores the junction of Management and Technology. Offered by the College of Engineering and Haas School of Business at UC Berkeley, it aims to launch ambitious innovators and is best suited for students with a passion for Science and a drive to find solutions through hands-on experience.

Dates: June 21 to July 8

Deadline: March 24

Fees: $ 10,499

Johns Hopkins Pre-College Programmes

Johns Hopkins University, The U.S.

Two-week college-level STEM courses in areas like Data science, Bioengineering, AI, and research fundamentals with flexible on-campus and online options. This is best suited to students wanting authentic university coursework and flexibility.

Dates: June 22 to July 3; July 6 to 17; and July 20 to 31

Deadline: March 10 and rolling thereafter

Fees: $6,140 (on-campus) and $1,950 (online)

Stanford Pre-Collegiate Summer Institutes

Online

A range of courses including Computer Science, Business, Math, Engineering, and Biosciences for students interested in deep and intense study with motivated peers.

Dates: June 15 to 26; July 6 to 17; and July 20 to 31

Deadline: March 13

Fees: $3,200

Tufts Pre-College Programmes

Tufts University, the U.S.

Various courses in Maths, Computer Science, Engineering, and Sciences. Meant for those seeking high academic rigour and flexibility of courses.

Dates: June 22 to 26; July 5 to 17; and July 19 to 31 based on the campus

Deadline: May 1 (Application); June 1 (Enrollment)

Fees: Varies by course

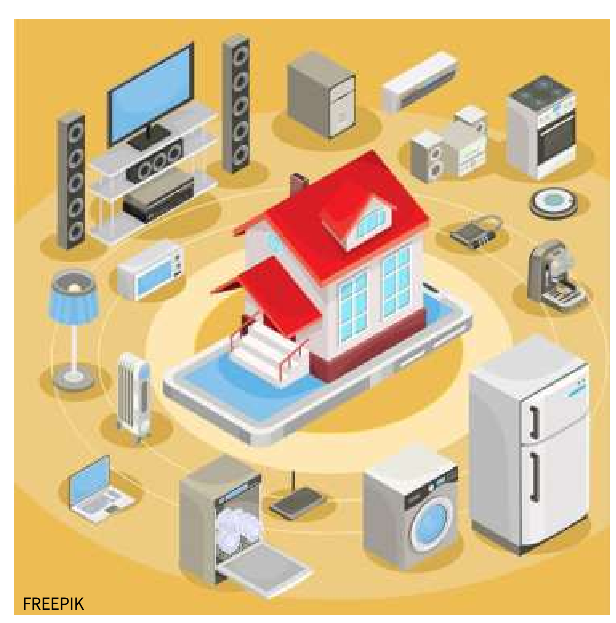
With inputs from Eeva Pachhapur

The writer is the founder and CEO of Inomi Learning, Gurugram. Email: info@inomi.in

Nalin Kumar

Homes redefined

As appliances grow smarter, quieter, and greener, the industry needs people to power its next leap


FREEPIK

In the last decade, kitchen appliances have gone beyond basic functions to offer comfort, efficiency, and modern design. Built-in hobs with safety sensors, quiet chimneys, precise ovens, and water-efficient dishwashers are increasingly common thanks to an industry that has been reinventing itself every year. This reinvention also brings a wave of opportunities for youngsters entering the world of manufacturing.

The home and kitchen appliance industry today is a modern ecosystem that involves engineering meeting automation, electronics meeting design, and sustainability meeting real-life problem-solving. Hence, the industry now seeks talent to keep pace with innovation, especially as companies are expanding into new categories of small and mid-sized appliances that require fresh design thinking, compact engineering, and feature-rich innovation.

Education

Students who wish for careers in this sector require solid academic underpinnings. Degrees in Mechanical, Electrical, Electronics, Mechatronics, Manufacturing, and Industrial Engineering will prepare graduates to work on the technical aspects of appliance development. Airflow dynamics, heat management, motor performance, and safety engineering are crucial to build high-performing appliances.

Advanced postgraduate programmes in Design Engineering, Thermal Systems, and Materials Science deepen this expertise and help one work on complex product engineering, whether improving motor efficiency, optimising heat flow, strengthening product structures, or designing less power-consuming appliances.

Diploma-level programmes are also critical. Technicians trained in wiring, calibration, moulding, and quality inspection play an important role on the shop floor and ensure that every appliance performs exactly the way it is designed to. Skill certification enhances their competitive edge. Programmes related to CAD design, automation systems, embedded electronics, IoT integration, and lean manufacturing help them keep pace with new-era production technologies and features.

Skills

Design and production capabilities increasingly incorporate advanced skills, such as Brushless Direct Current (BLDC) motor technologies, energy-efficient heating systems, advanced sensor integration, and smart and connected digital interfaces. Grasping how these systems interact – whether in chimneys, ovens, dishwashers, or a new generation of compact kitchen appliances – enables engineers to contribute directly to product innovation.

Modern manufacturing practices are also changing. Automation, robotic assembly, data-driven quality checks, and high-precision tooling require professionals who can work hand-in-hand with technology. Reliability engineering is becoming important, as consumers expect appliances to work faultlessly for years, making long-term durability testing and safety compliance crucial to a company's reputation. Today, sustainability lies at the heart of every design conversation. Minimising power consumption, improving material recyclability, and creating water-efficient systems are essential considerations that shape the industry's future.

Careers

The industry offers a wide spectrum of career roles. Product design and R&D attract those who enjoy creative problem-solving. Manufacturing and operations roles suit those who like working on process improvement and efficiency. Quality and testing roles appeal to analytical professionals who take pride in ensuring reliability. Supply-chain and vendor-development roles continue to grow as companies scale. Service and customer-experience teams are equally important, especially as appliances become more feature-rich and require specialised support. Engineers who blend technical knowledge with market understanding transition into product-management roles, where they guide product strategy and long-term road maps.

As homes expand, expectations rise and technology continues to reshape the appliances sector, youngsters have enormous opportunities to innovate and redefine how families find comfort every day.

The writer is CEO of Kaff Appliances.

The AI-ready clinician

What students need to know about AI in hospitals

Rustom Lawyer

FREEPIK

Walk into any modern hospital today and you will encounter AI working silently across multiple domains. Clinical decision support systems analyse vast patient datasets, medical histories, and current symptoms to provide evidence-based recommendations in real-time. Diagnostic tools detect conditions from medical imaging with remarkable precision, while AI-enabled wearable devices monitor vital signs continuously, allowing immediate intervention when irregularities arise.

AI-powered systems are revolutionising patient flow and resource allocation. Workflow automation handles scheduling and triage prioritisation, freeing professionals to focus on what requires human judgment. Ambient clinical intelligence is transforming documentation, capturing natural conversations between doctors and patients, and automatically structuring them into clinical notes.

Therefore, understanding what AI can and cannot do in the clinical environment is essential. Core competence will lie in collaborating intelligently and treating AI as a powerful, but fallible, colleague. Understanding how medical data is collected, cleaned, and labelled is essential because biased or poor-quality data translates directly into unsafe or unfair recommendations. Three clusters of skills are especially important:

AI and data literacy: Understanding how machine-learning models are trained, validated and monitored, and what sensitivity, specificity, false positives and false negatives mean.

Ethics, law and privacy: Consent for data use, confidentiality in a cloud-enabled world, and the practical implications of frameworks such as Health Insurance Portability and Accountability Act (HIPAA), General Data Protection Regulation (GDPR), and emerging data-protection norms.

Medical data handling: Safe documentation practices, de-identification basics, and disciplined use of electronic medical records.

Students should learn to work in interdisciplinary teams, translating clinical realities into problem statements for engineers and explaining technical trade-offs back to colleagues and patients.

New careers

The convergence of AI and healthcare is creating new career pathways. Healthcare data scientists analyse clinical systems data; clinical informaticists build interfaces for clinical handovers; AI research scientists design predictive models for diseases or image recognition tasks, and bioinformatics analysts run genomic-driven personalised treatments.

Trust remains the primary barrier to AI adoption. Regulatory frameworks weren't designed for technologies that continuously learn and evolve. Algorithms trained mostly on data from specific geographies or socio-economic groups can misdiagnose or under-serve others. Automation bias is a concern. Once an AI tool is in the workflow, humans can become overly inclined to accept its suggestion, even when clinical judgement or the patient's story points elsewhere. Other key areas include:

Privacy and security: Health data is among the most sensitive.

Workforce impact: Effective automation can remove drudgery while poorly designed automation can deskill clinicians.

Trust and transparency: Patients and clinicians need an understanding of how a system reached its recommendation.

AI must augment human expertise, never replace it. Students entering healthcare today have a rare opportunity to shape how AI is used in hospitals. Those who combine strong clinical foundations with ethical, data-savvy engagement will help design a health system that is more intelligent and humane.

The writer is Co-Founder and CEO, Augnito.